

openmaterials.ai

A database for physics.

The schema is the physics itself.

The same physics, a dozen codes

kaldo · phono3py · ShengBTE · LAMMPS · VASP · Quantum ESPRESSO

They rarely agree, and published computational results rarely reproduce.

The workflow lives in scripts and tribal knowledge.

Store the physics, not the printout

A graph of typed **variables**, connected by the **formulas** that relate them.
The law itself, not a number in a table.

$$\kappa^{\alpha\beta} = \frac{1}{V} \sum_{q\nu} c_{q\nu} v_{q\nu}^{\alpha} v_{q\nu}^{\beta} \tau_{q\nu}$$

Every code maps onto the layer

Seven adapters pin each code's outputs to the variables, with units and conventions. Variables mapped per code:

kaldo · 32

phono3py · 31

ShengBTE · 20

phonopy · 17

GPUMD · 8

LAMMPS · 8

ASE · 1

Simulations and experiments attach

Values from codes (**numerical**) and papers (**lab**) attach to the same variables and reconcile at the observable.

Silicon thermal conductivity, Tersoff, 8×8×8, 300 K

RTA **kaldo** 19.46 · **phono3py** 16.74

direct **kaldo** 26.91 · **phono3py** 24.30 W/m·K

A computable, self-checking database

Because the formulas are stored too, it can

derive · reconcile · invert

run the formulas forward, cross-check redundant routes, and (next)
solve them backwards. Not a data lake.

Open and forkable

The whole database is files in a repo.

`git clone` to run locally · pull request to append

Storing science the way git stores code.

A semantic layer for scientific AI

Agents act on typed physics, not text. An invalid step is a **type error**, not a silent wrong number. Redundant routes are a built-in verifier, so verification scales with compute.

The standard to store science.

[explore the live map](#) → · [openmaterials.ai](#)

Speaker notes